Topic 17: On-Chip Variation

ECE 9433: Agile System-on-Chip Design

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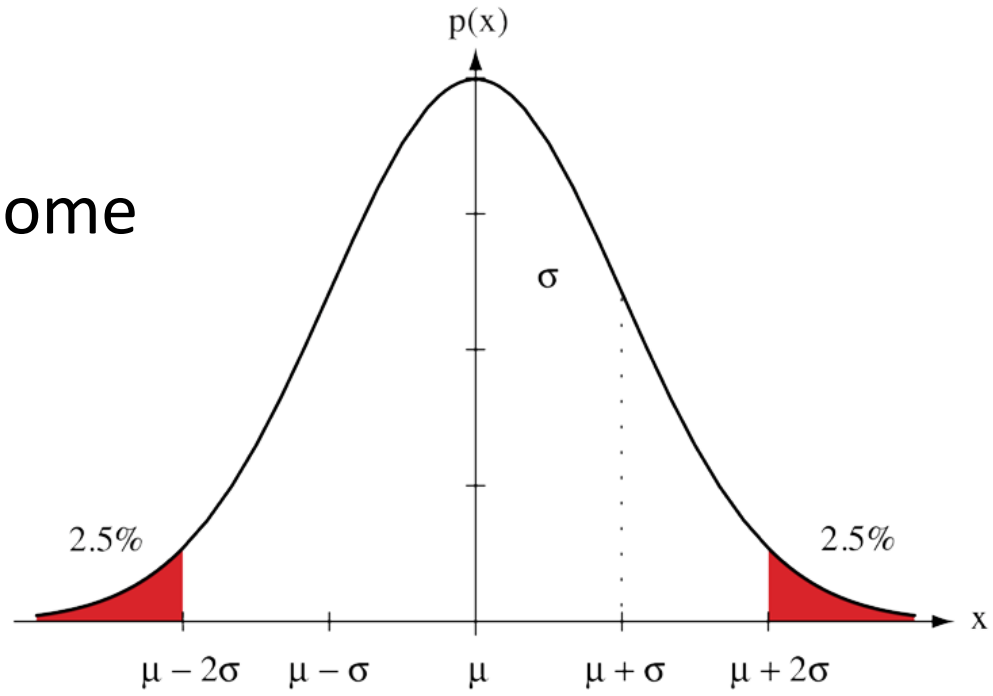
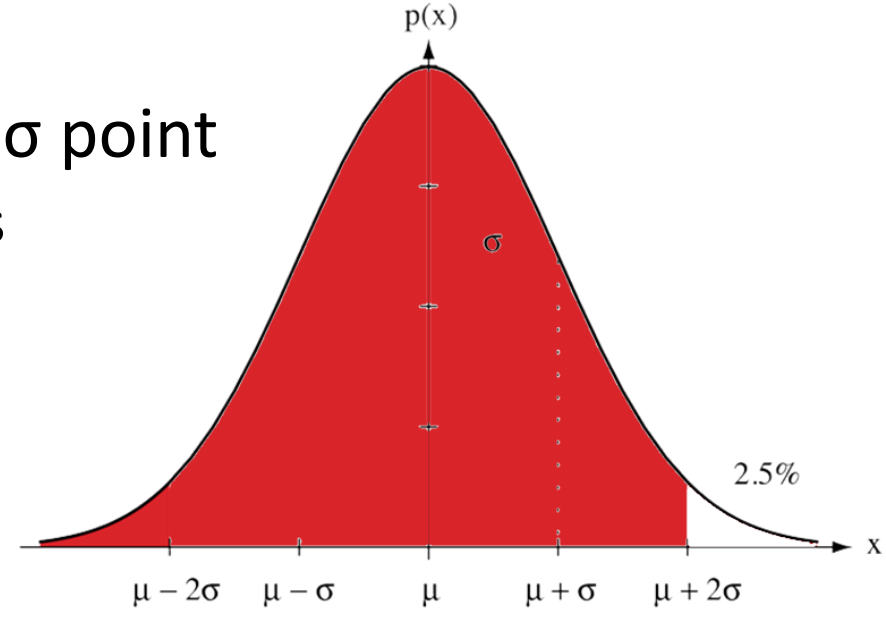
Variation

- Any real process is inherently imprecise
- Every value in the chip design will have some variation, usually modeled as a normal distribution
- Main kinds of variation modeling:
  - Process corner modeling
  - Random sample modeling (Monte Carlo)
  - Statistical modeling
- Two main processes to model
  - FEOL (transistors)
  - BEOL (metals)



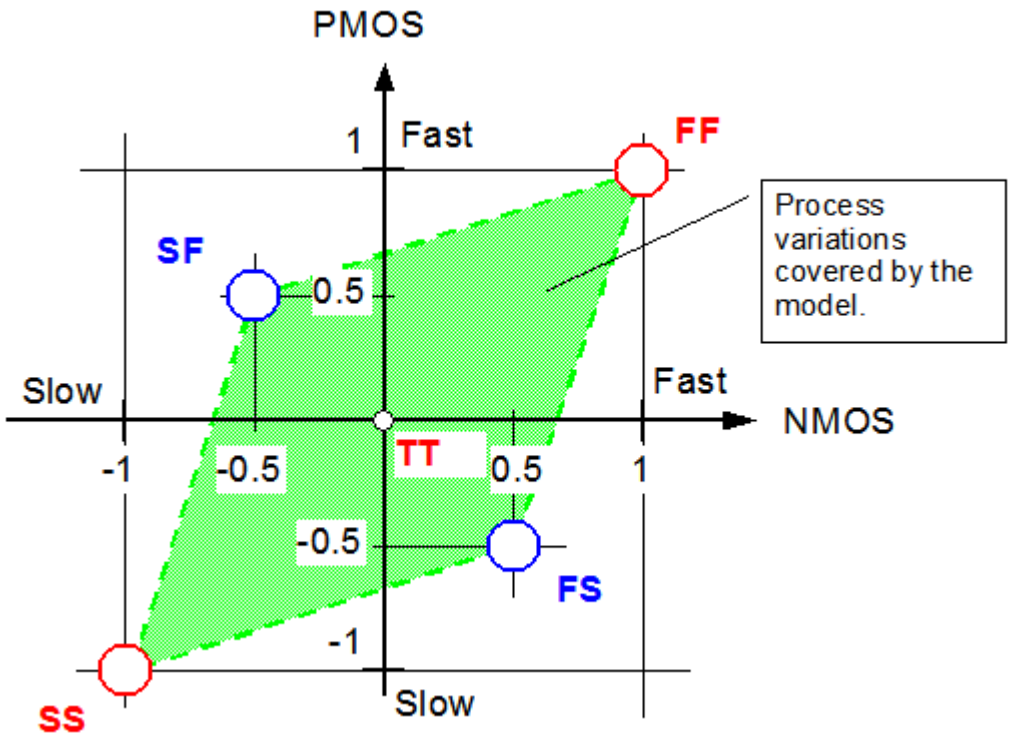
Process Corner Modeling

- Pick a point on the distribution and assume that all structures fall on that point
- Example: Assume that all structures are exactly the mean value (μ)
  - This is how you've been working up until now
- Example: Assume that all structures are the +2σ point
  - This will model the worst case for 97.5% of devices



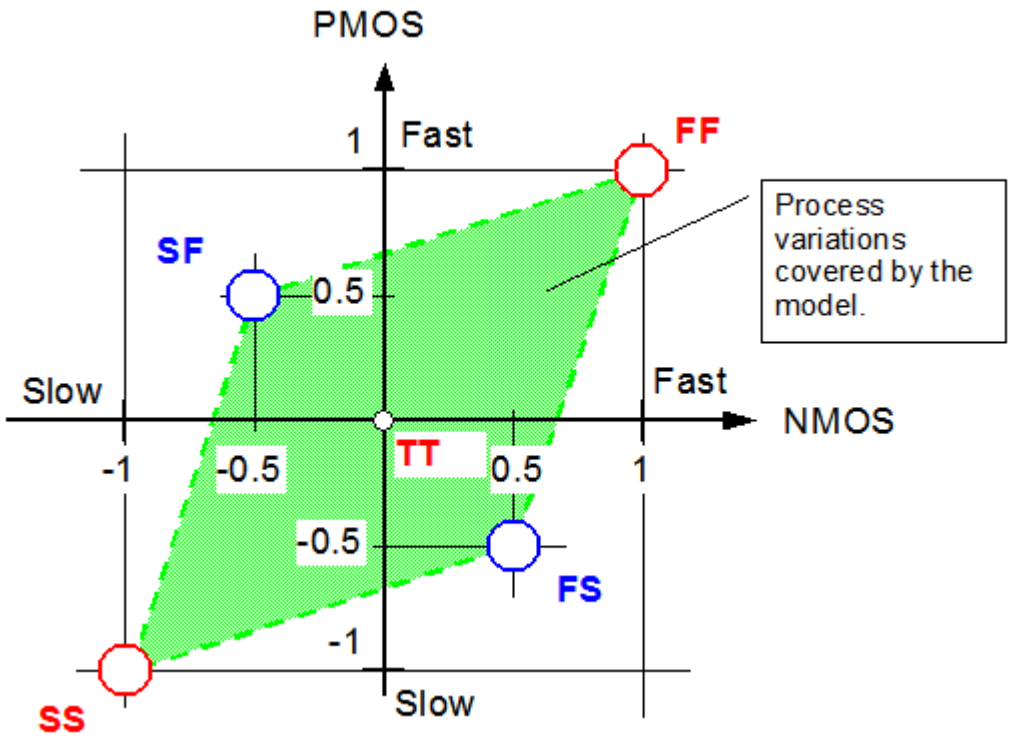
FEOL Corner Modeling

- Goal: model all “worst cases” for transistors
- NMOS / PMOS can have different distributions = 2 dimensions to sample
- Pick “corners” of the distribution
  - One end of distro. is “fast”, other is “slow”
  - NMOS/PMOS corners: SS, SF, FS, FF
  - Additionally, TT for average case of circuit
- Most PDKs use ±2-3σ for “worst case”



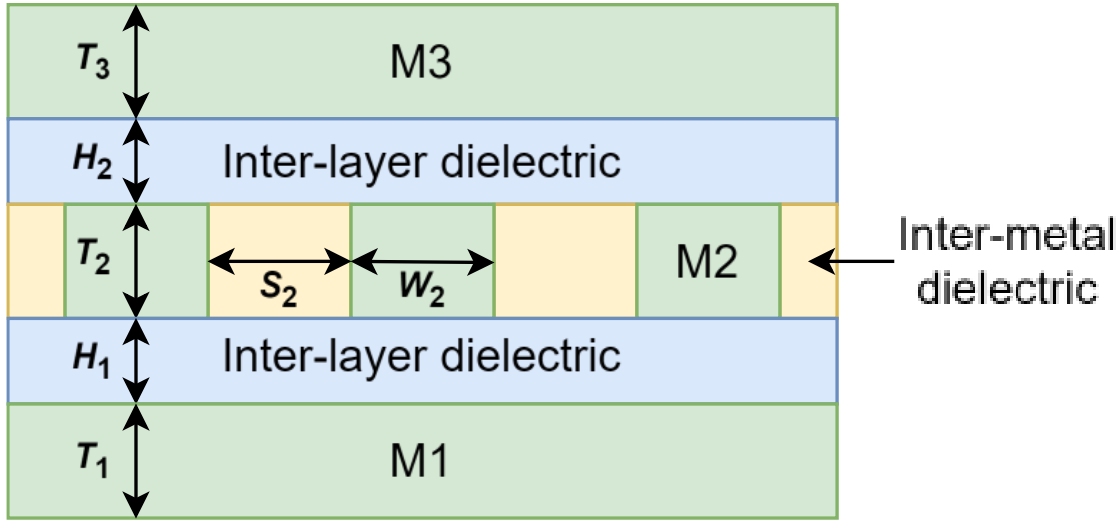
FEOL Corner Modeling

- Modeled variation sources:
  - Doping concentration of substrate
  - Channel length
  - Channel width
  - Oxide thickness
- Other variation sources:
  - Imperfections / defects



BEOL Corner Modeling

- Goal: model all "worst cases" for wires and vias
- How to model? 3 main parameters:
  - ΔW: change in wire width
  - ΔT: change in wire thickness
  - ΔH: change in inter-layer dielectric
- $t_d = RC$
- $\uparrow \Delta W = \uparrow C, \downarrow R$
- $\uparrow \Delta T = \uparrow C, \downarrow R$
- $\uparrow \Delta H = \downarrow C$
- So which combination is the worst case?



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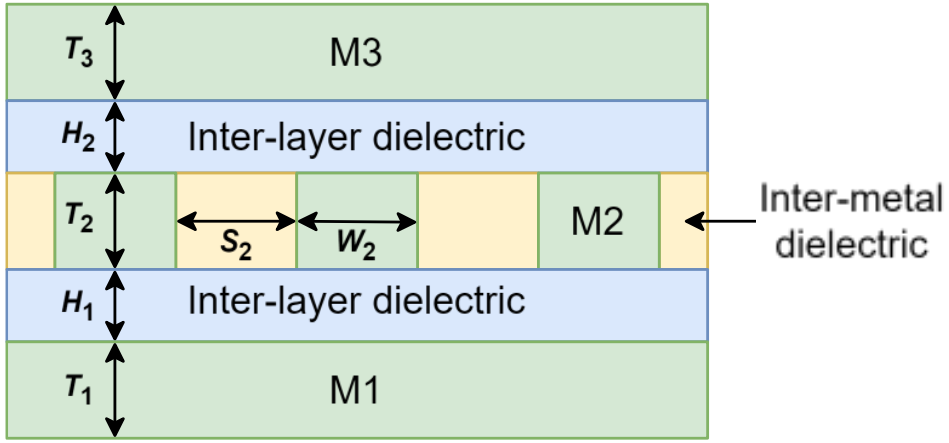
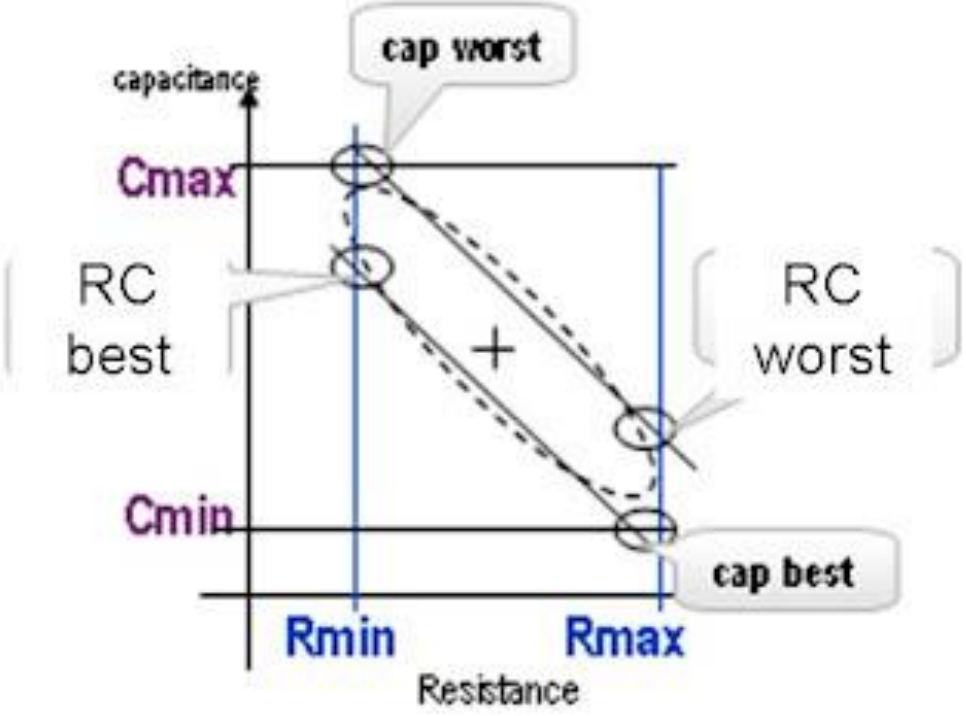
BEOL Corner Modeling



- Worst depends!
- Capacitance dominates for shorter wires
- Resistance dominates for longer wires
- 5 main corners (usually at ±2-3σ):

	ΔW	ΔT	ΔH
Typical	typical	typical	typical
C-best	min	min	max
C-worst	max	max	min
RC-best	max	max	max
RC-worst	min	min	min



Additional Corner Modeling

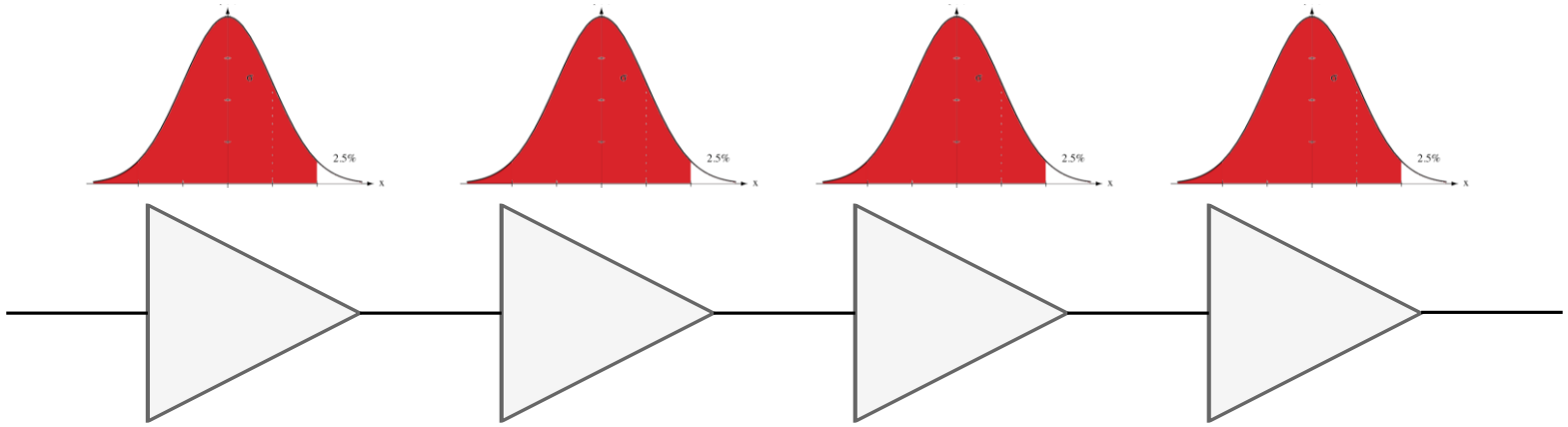
- Need additional characterization points for:
  - Each operating temperature
    - Usual characterization points: -40C, 20C, 80C
  - Each operating voltage
    - Usual characterization points: nominal - 10%, nominal, nominal + 10%
- Unusual characterization points may need to be compiled by the designer (i.e. from SPICE model into .lib)
- Very unusual characterization points may not even be provided by the foundry
  - In this case, you would have to pay someone to do it or do it on your own

Skew Wafers

- For large production volumes, you may want to test the extremes of the process to ensure your design is resilient
  - But the extremes inherently happen in a very small % of the chips!
- Solution: many foundries offer “skewed” processes where the equipment is adjusted to produce ±2-3σ sample points as the mean
- This provides many samples to enable reliability testing

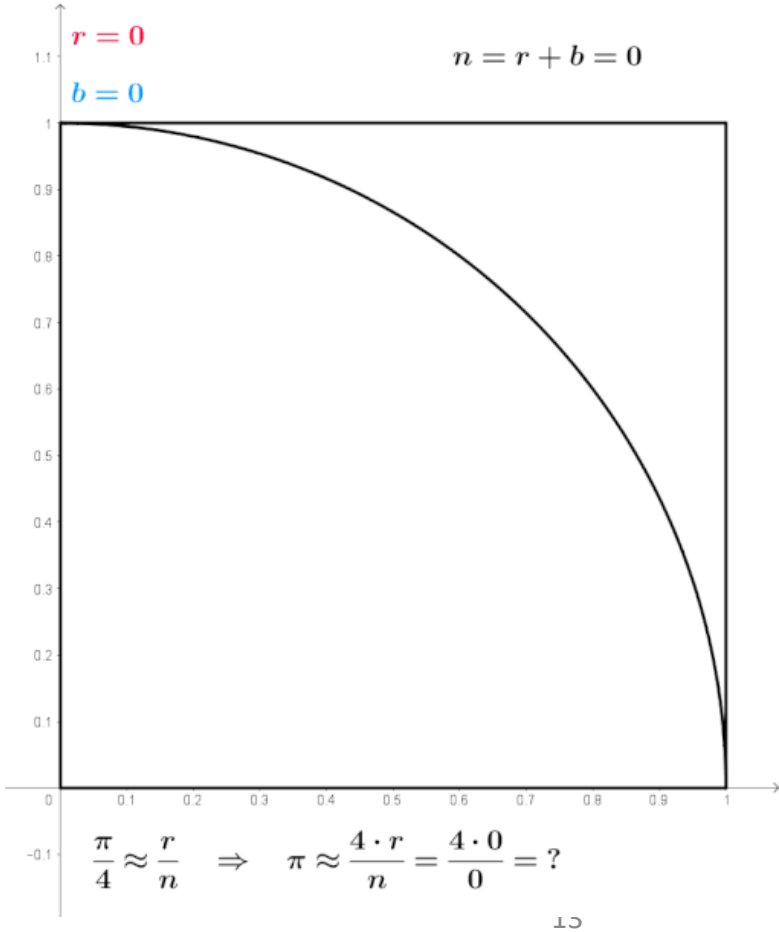
Pros/Cons Corner Modeling

- Can be fast to compute with a small number of corners
- “The corner explosion”: each additional characterization criterion leads to exponential increase in corners
- Can be overly pessimistic
  - Every device being at ±2-3σ means that there is near 100% sample correlation, which is usually not the case



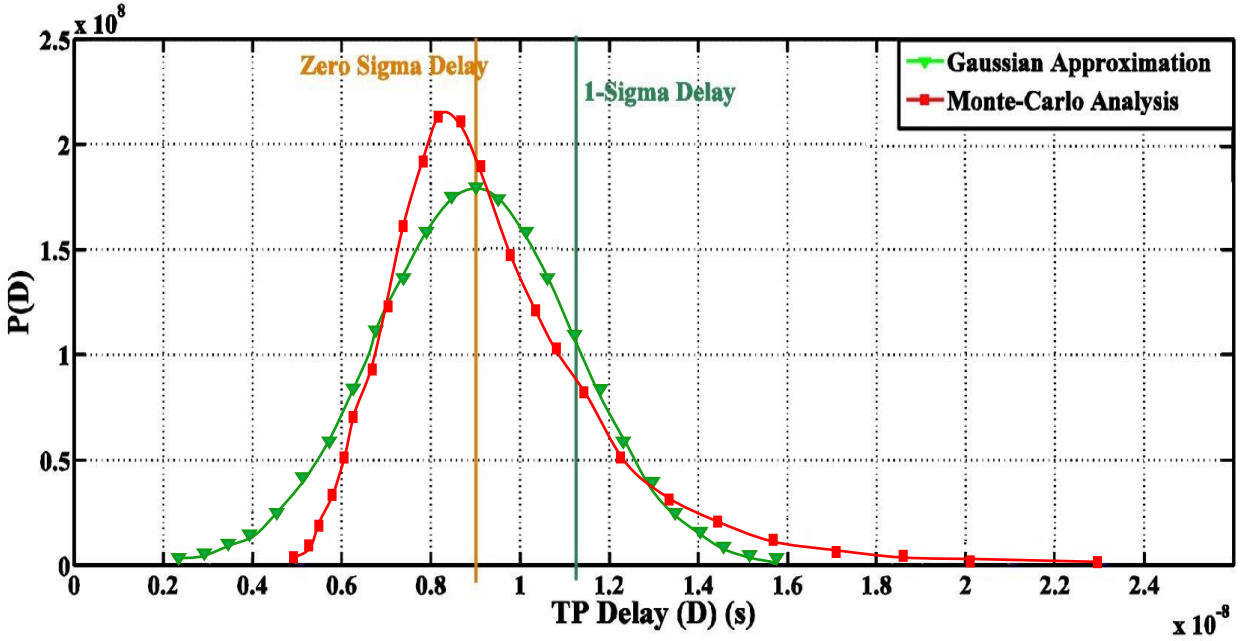
Statistical Sampling (Monte Carlo)

- Monte Carlo method: accumulate random samples until you can approximate a solution (within a bound of confidence)
- For circuits: pick a random sample for each gate parameter and then do analysis (i.e. STA) with it



Statistical Sampling (Monte Carlo)

- Monte Carlo is gold standard of accuracy in modeling variation, but it is *very slow*
- If we can make some assumptions about the distribution, we can speed up the calculations



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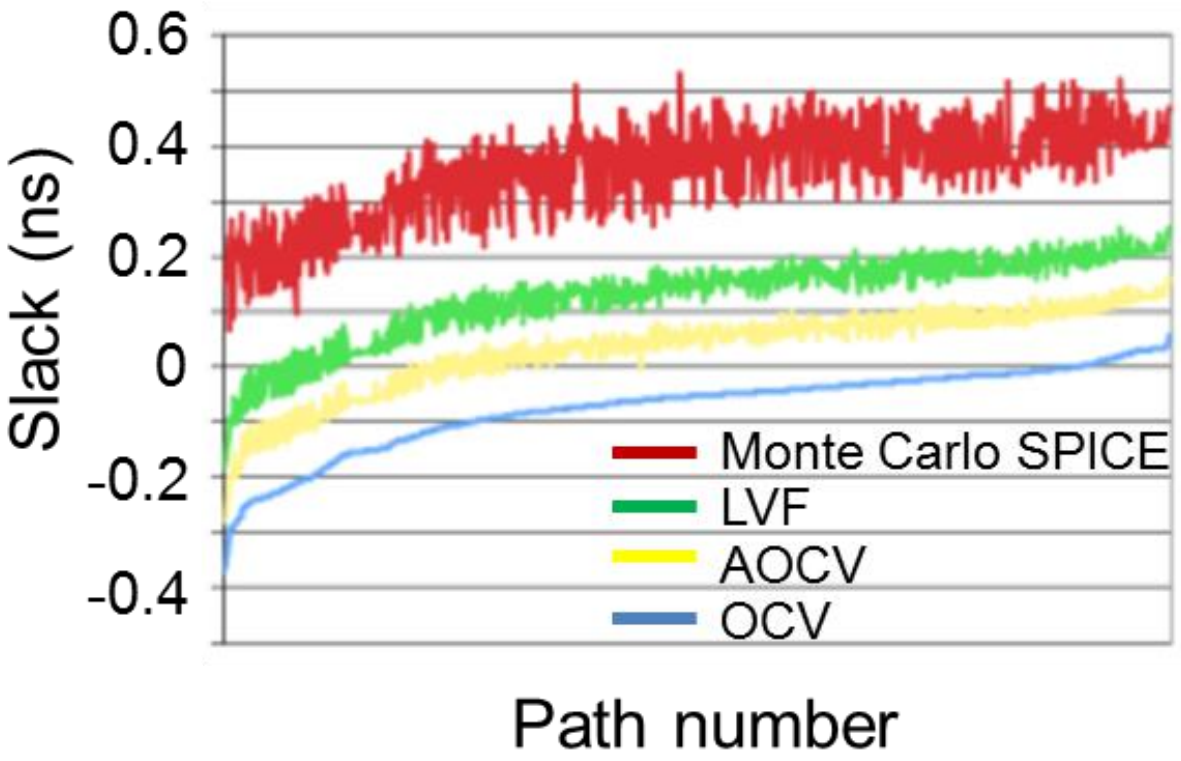
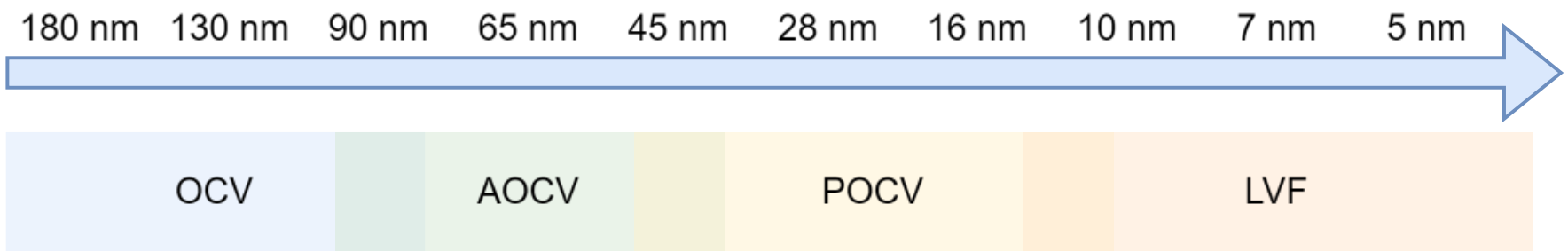
Statistical STA

- Represent the distribution as a Gaussian and characterize each gate with μ and σ parameters
- STA: scalar input, scalar output
- SSTA: distribution input, distribution output
- Much faster than Monte Carlo, with some losses in accuracy



Implementations of Variation Modeling

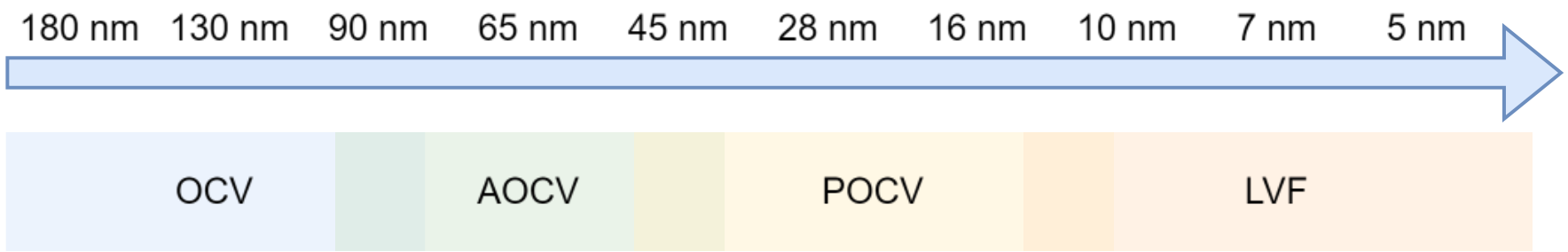
- **OCV**: On-Chip Variation
- **AOCV**: Advanced On-Chip Variation
- **POCV** (SOCV): Parameterized (Statistical) On-Chip Variation
- **LVF**: Liberty Variance Format



*Not precise

On-Chip Variation

- Mostly used in very old nodes (90 nm+)
- Relies on simple derating set in your SDC file
  - Multiply your gate timing by a derating factor, e.g. 10%
- Very pessimistic modeling



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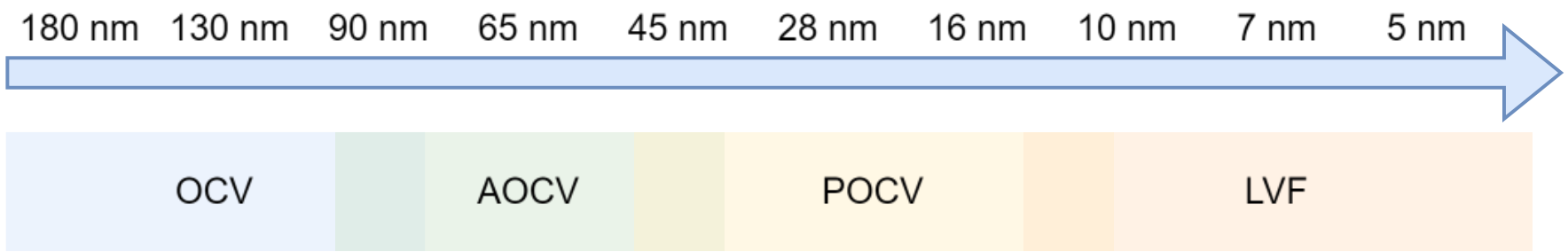
Advanced On-Chip Variation

- Mostly used in older nodes (~90 nm to ~45 nm)
- Uses a distance-base variation heuristic
  - Assume that gate variation is correlated based on how close they are together
  - Much better than a flat derating factor across all gates
- Still somewhat pessimistic modeling



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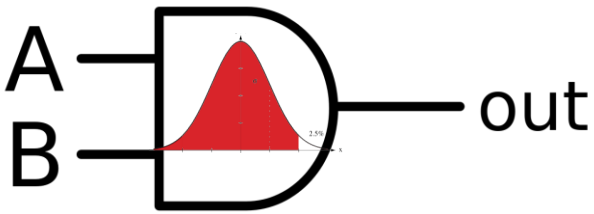
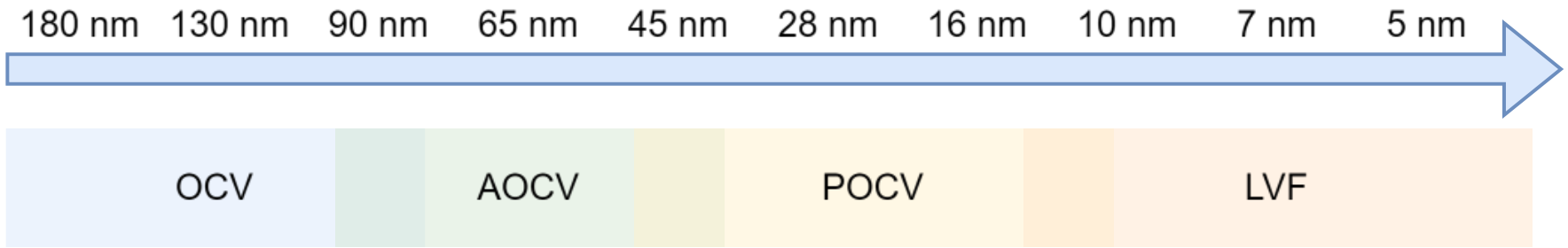
Parametric On-Chip Variation



- Mostly used in newer nodes (~45 nm to ~16 nm)
- Use statistical modeling per cell, and perform statistical STA
  - You can set your own desired sigma for your corners
  - Better than AOCV, much less pessimistic



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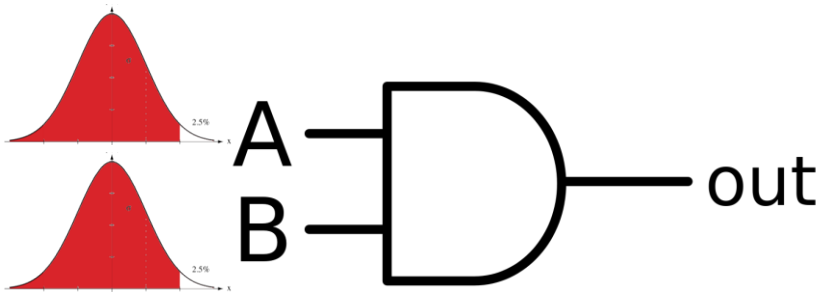
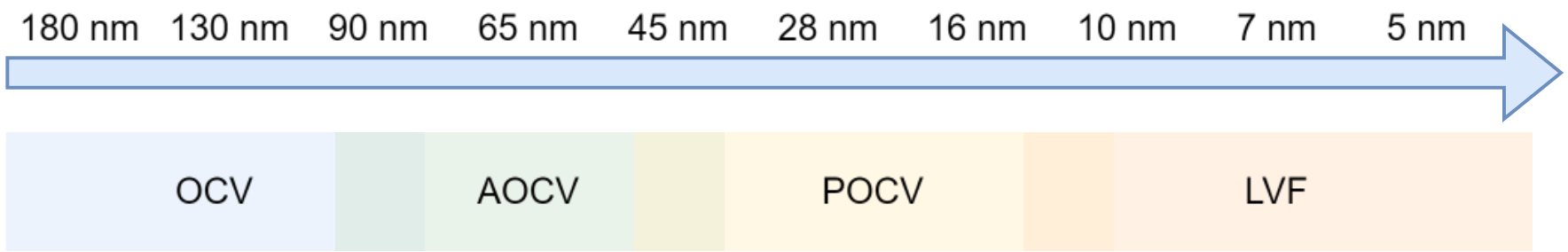
Liberty Variance Format



- Used in advanced nodes (~16 nm and under)
- Use statistical modeling per timing arc (each input and output transition) and perform statistical STA
  - You can set your own desired sigma for your corners
  - Basically the same model as POCV, but applied to every gate transition, not just for the whole cell



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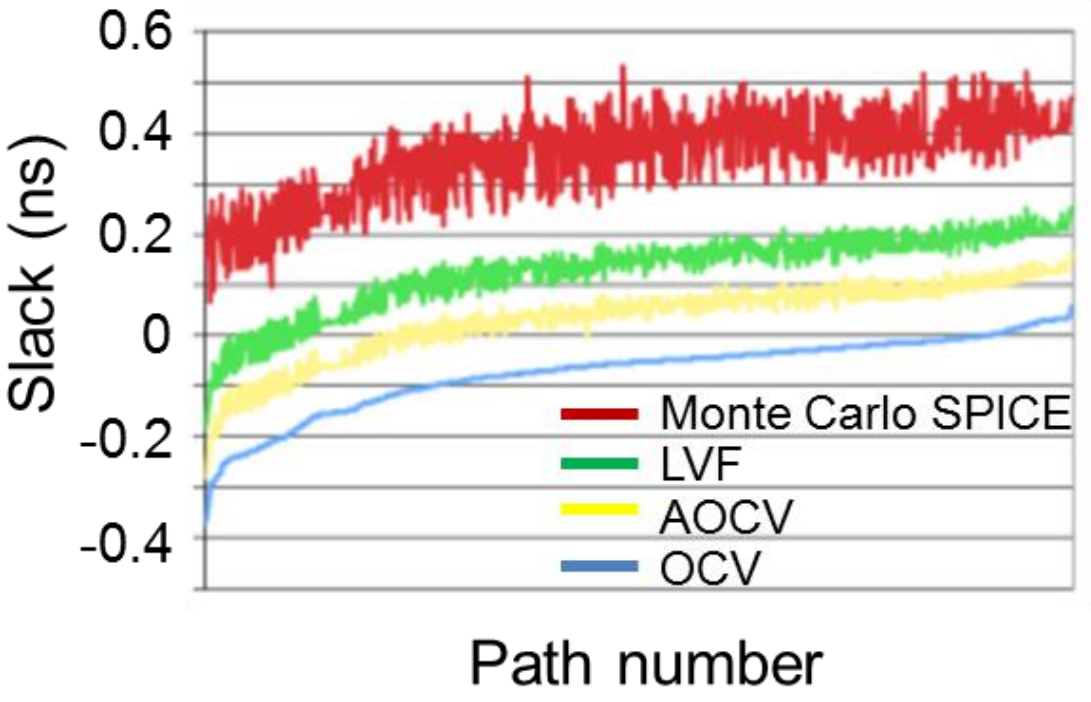
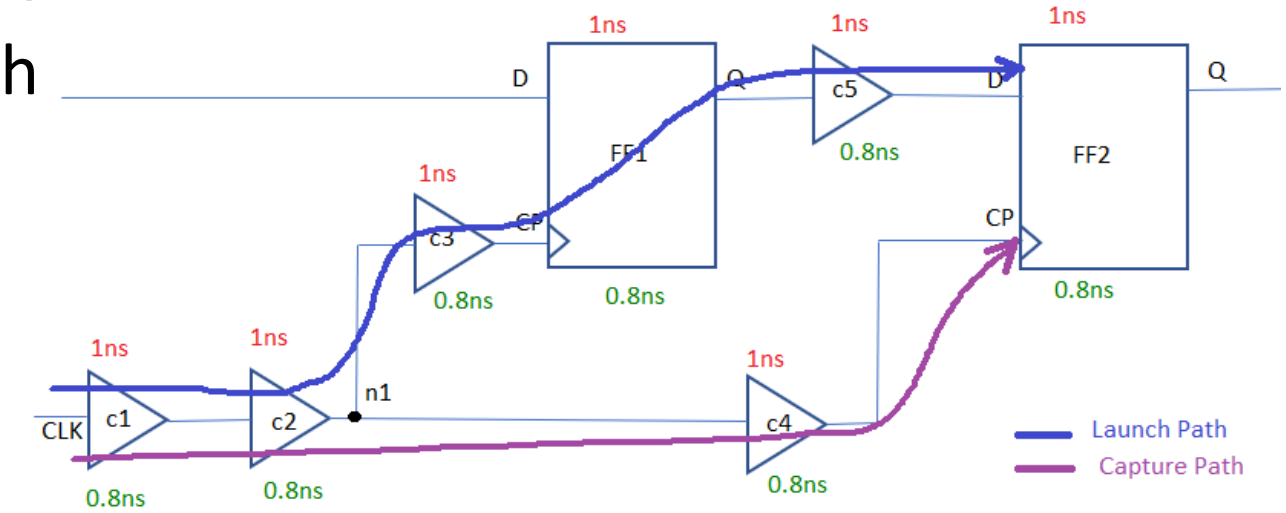
OCV Comparison

- As always, more accuracy costs more runtime!
- Note: advanced OCV models work on older nodes; the problem is that the manufacturer hasn't characterized those models for you



Aside: CRPR

- When modeling variation, especially in clock paths, two separate analyses of paths may give overly pessimistic timing
- A buffer cannot be simultaneously best and worst!
- **Clock Reconvergence Pessimism Removal (CRPR)**: remove difference in timing on shared timing path segments
- Example: c1 and c2 cannot be both 0.8ns and 1ns
- Remove (1 - 0.8) = 0.2 ns of pessimism



Multi-Corner Constraints

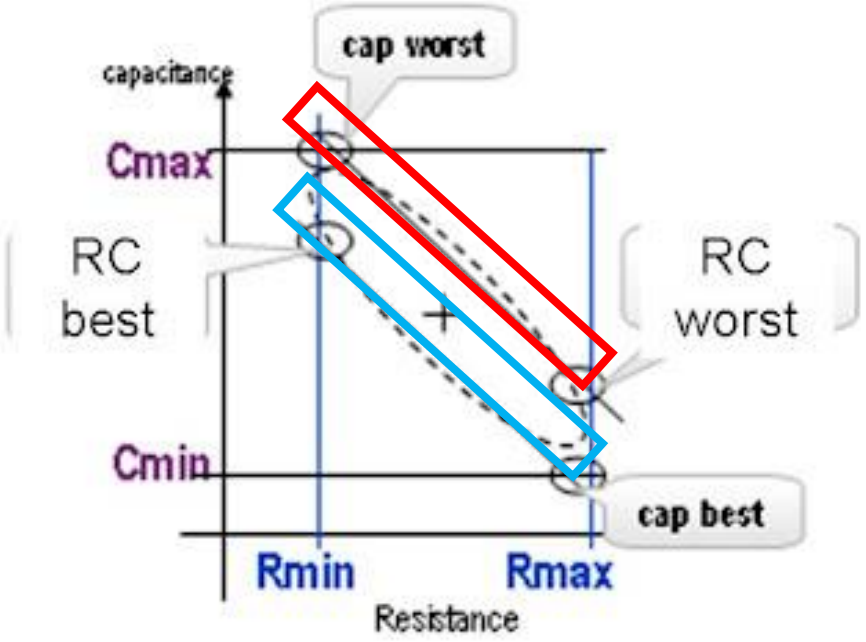
- How do we analyze our chip with multiple corners (sigmas)?
  - We don't necessarily need or want the chip to run the same at a slow corner vs. fast corner
- Create **analysis views** to represent how to view the cell
- An analysis view is a set of operating conditions and constraints the tools will use to perform STA
  - Example analysis views:

View	VDD	Temp	FEOL	BEOL	Clock
Slow	90% nominal	80 C	SS	RC-worst	600 MHz
Fast1	110% nominal	-40 C	FF	RC-best	800 MHz
Fast2	110% nominal	80 C	FF	RC-best	800 MHz

Multi-Corner Constraints

- Analysis views must be marked as **setup** or **hold**
  - You will have to know whether the corner represents the fastest timing (hold) or slowest timing (setup)
- Analysis views are used during optimization to ensure setup and hold are met
- Some tools also have **report views** which do not impact optimization, but will provide timing reports for that corner

Slowest corners should be used for setup checks

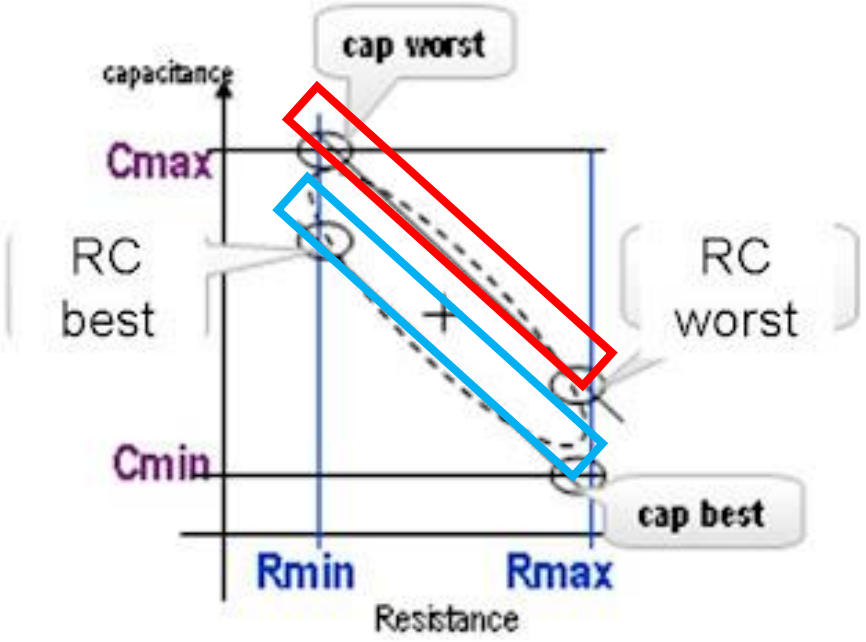


Fastest corners should be used for hold checks

Practical Multi-Corner Constraints

- Runtime can slow quickly with more corners
- For prototyping, when performance is usually an optimization target rather than a constraint, you don't need to optimize setup for the fast corner
- You can use typical for setup, fast for hold
- Use one corner for each (or even 1 corner overall) and then add corners as design approaches finalization
  - Same approach as CCS vs. NLDM modeling

Slowest corners should be used for setup checks



Fastest corners should be used for hold checks

Multi-Mode Constraints

- You can have different “modes” you test your chip in
- For example, a factory test mode for debugging, and an operational mode
- You can set SDC constraints for each mode and have them analyzed separately
- Used in industry
- Not used often in prototyping

Acknowledgement

- Kahng, Andrew B. Design Automation Conference, Session 12: Timing Closure. 2015. “New Game, New Goal Posts: A Recent History of Timing Closure.” 2015. (Presentation)